

14 Hydrocarbons

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Alkanes undergo free-radical substitution with halogens under UV light, often producing mixtures. Alkenes undergo electrophilic addition because π electrons attract electrophiles. Benzene is stabilised by delocalised π electrons and typically substitutes rather than adds, preserving aromaticity.

Worked example: follow the reasoning

Ethene decolourises bromine water: Br_2 adds across $\text{C}=\text{C}$ to form 1,2-dibromoethane.

Quick check · multiple choice

A Which reaction class describes HBr addition to propene?

[1]

- A Free-radical substitution
- B Electrophilic addition
- C Elimination
- D Condensation

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 State the mechanism for adding HBr to propene and predict major product using carbocation stability.

[3]

Answers and explanation

Quick check: B. Electrophilic addition

Question 1 [3 marks]

Electrophilic addition [1]; H^+ adds to terminal carbon to form more stable secondary carbocation [1]; major product 2-bromopropane [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.